

KEA HARVAN SUYANTO

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TECHNICAL SKILLS

- Languages: Python, SQL, Java, C++, C, JavaScript
- Generative AI: LangChain, LlamaIndex, Google ADK, Promptfoo, Ragas
- Data/Machine Learning: Apache Airflow and Spark, PyTorch, Google OR-Tools
- DevOps: AWS, Docker

WORK EXPERIENCE

Data Scientist / ML Engineer Intern, Starhub Ltd, Singapore

May 2025 - Nov 2025

- Built an agentic regression test harness (LangGraph/LangChain) for [Agent Atlas](#) that generates and validates 10+ synthetic tool-call test cases per run, catching tool-output regressions before release via automated assertions and benchmarks
- Improved [Agent Atlas](#) tool-calling accuracy by 30%+ by integrating Model Context Protocol (MCP) and refactoring tool schemas/response parsing to reduce malformed calls on complex customer queries
- Implemented an Airflow DAG pipeline that converts processed TelCo user data into structured narrative + chart metadata using LLMs, enabling fully automated dashboard generation for the [Smart Retail Platform](#)
- Engineered and deployed a Mixed Integer Linear Programming (MILP) optimisation engine for the [Smart Retail Platform](#) using OR-Tools and Apache Airflow; Utilised TelCo data to inform client decisions on advertising asset placements, achieving globally optimal results

Lab Tutor, National University of Singapore, Singapore

Jan 2025 - May 2025

- Conducted weekly 2-hour learning sessions for 30+ students on the Java programming language, focusing on programming paradigms such as Object Oriented Programming and Functional Programming
- Cut grading turnaround for 150+ Java assignments by automating compilation, test execution, and result summarization with Bash scripts and deterministic test fixtures

AI Engineer Intern, Pt. Adi Citra Sakti, Semarang

Dec 2024 - Jan 2025

- Trained and containerised a facial recognition model capable of processing multiple video streams using OpenCV, Docker and image augmentation techniques, achieving a precision of 70% via hyperparameter tuning, ensuring accurate identification across concurrent real-time inputs
- Built a standardised evaluation framework utilising confusion matrices to audit the multi-class multi-label model, streamlining the testing pipeline to reduce manual analysis time, ensuring robust validation prior to deployment

EDUCATION

National University of Singapore

Aug 2023 - Present

Bachelor of Computing in Computer Science

- Focus: Artificial Intelligence and Database Systems, with an interest in ML Systems Design (Expected Graduation Dec 2026)

Singapore Management University

Aug 2022 - Jun 2023

Bachelor of Business Management

- Awarded Dean's List for the Lee Kong Chian School of Business for Academic Year 22/23

PROJECTS

[Comparative RAG Framework for Trade Classification](#), Singapore

Feb 2026 - Present

- Built an automated RAG evaluation harness over Singapore trade regulation docs using LlamaIndex and Ragas/Promptfoo, running matrix tests across 4+ LLM providers and prompt variants to compare retrieval accuracy, latency, and cost/query

[Agentic Mobile App Regression Tester \(TikTok Hackathon\)](#), Singapore

Aug 2025 - Aug 2025

- Implemented an agentic mobile UI regression tester that translates natural-language test steps into executable UI actions using an MLLM + Google ADK, with Pydantic-validated action schemas to reduce invalid interactions
- Designed an end-to-end testing pipeline, capturing in-progress screenshots, feeding them into a Computer Vision (CV) system utilising OpenCV for image difference detection, and generating detailed regression reports